Reassessing the ARRA Employment Multiplier: A Continuous Treatment Approach

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Abstract

The following paper has the objective of completing a comprehensive reassessment of the effects on employment that the ARRA spending policy carried across the fifty United States between February 2009 and December 2010. It does so by implementing modern estimation techniques to the existing framework and evaluating the results previously obtained through a careful interrogation of its inner assumptions. By exploiting the panel data structure of the observations these models are able to account for the evolutionary and compounding effects of the job multiplier. Such novel application yields new insights into the employment effects caused by investment policies in the aftermath of economic crises and sets a new potential framework in which such effects can be estimated.

I Introduction

The American Recovery and Reinvestment Act (ARRA) was a stimulus package enacted in February 2009 to counteract the negative effects that the Great Recession had been causing to the American labor market. When the act was passed, it was expected to inject in the American economy a total of $787 billion and it overall accounted for a spending of over $831 billion, according to the Congressional Budget Office (2012). The stimulus package was largely concentrated in three public investment areas: healthcare, education and infrastructure. ARRA payments were divided into three main categories: announcements, obligations and actual outlays. Conditional on authorization by the ARRA legislations, funds were allocated to different federal agencies. These funds were later announced as available to the eligible recipients within each state and afterwards awarded to them. A federal agency is said to be obliged to the recipients that are entitled to receive the payments after the awarding. The subsequent and last step is the actual payment, or outlay, of the funds. Building upon the existing literature (Wilson, 2012; Feyrer and Sacerdote, 2011; Chodorow-Reich, 2019) I instrument for the cumulative spending received by each state, in the form of actual outlays, on statutory formulas pre-existing to ARRA announcement. Such instruments are split in the three mentioned areas of investment and exploit equations that, based on structural and unvarying characteristics such as miles of highways, school-aged population or Medicaid expenditures, determined the amounts of federal investment in each state prior to ARRA. This is done to remove potential endogeneity deriving from state administration quality and to get the proportion of investments approved due to structural characteristics rather than Governor ability at soliciting funds or proposing better projects. Dissociating from the existing literature that operates a cross-sectional 2SLS estimation after the implementation of additional controls, this paper uses these first-stage fitted values to construct treatment and control group and then apply two difference-in-difference techniques to estimate fiscal job multiplier. Such implementation differs substantially from previously realized analysis as it is able to i) control for within-state fixed effects ii) control for between-state time fixed effects iii) study the monthly evolution of ARRA spending effects iv) study the heterogeneous effect of ARRA spending across dosage groups. The tools used in this paper provide a solid contribution to the literature as they are able to extend the cross-sectional analysis to a dynamic study able to account for heterogeneous effects. While largely diverging from the earliest results found in this literature, the results reported in this paper are essentially aligned with the most recent development and tend to estimate a cost per job created or saved of approximately $50,000.

II Data

The main source of data used to the purpose of this paper is the harmonized dataset created by Chodorow-Reich (2019). It comprehends a collection of monthly state-level

data spanning from December 2007 up to January 2011. It contains key variables coming from previous estimations such as statutory formulas for highway and education funding (Wilson, 2012) or congressional seniority (Feyrer and Sacerdote, 2011). On top of that are added relevant variables not available at the time early studies were completed such as gross-state product (GSP) and a new standardized form of created job counting. To this extent, the Chodorow-Reich (2019) dataset is considered the most up-to-date harmonized dataset on ARRA spending. The primary source for all these observations is the Recovery.gov website containing granular statistics at the state-level reported by federal agencies. The Bureau of Labor Statistics (BLS) provides evidence related to current employment statistics (CES), the main output variable of interest, and quarterly census of employment and wages (QCEW). Additional macro-indicative data is extracted from the Bureau of Economic Analysis (BEA).

III Econometric Analysis

1. Baseline Specifications

The core idea behind a difference-in-difference study is centered around the evolutionary study of two groups of observations: the treatment and the control group. In randomized control trials (RCT) these two groups are equally sampled from a population and, for enough large samples, they are assumed to be exact equal representations of the population of interest. This fundamental assumption allows for the isolation and identification of the treatment effect once the treatment is assigned to only one of the two groups. The treatment itself must be the only variable that distinguishes the two cohorts and thus the exclusive factor that impact a subsequent evolution of the variable of interest. The application of this technique to real-world observable data is feasible only upon the rigorous compliance to some assumptions: unconfoundedness, parallel trends assumption, no anticipation and stable unit treatment value assumption (SUTVA). The unbiased estimation of the average treatment effect on the treated (ATT) thus closely depends on i) whether there are no confounding variables that endogenously determine which group gets the treatment, ii) if the two groups' variable of interest follow the same trajectories before the treatment, iii) whether the effect of the treatment was not anticipated before the assignment and iv) if there are no spillovers between the treated and the control group. Since, to the best of my knowledge, the application of difference-in-difference estimation is completely novel to the analysis of the ARRA effects on employment, I will start by discussing the validity of these assumptions before describing the specifications implemented.

The first important point to mention is that the structure offered by the data and the procedure followed for the distribution of the ARRA payments does not allow for a simple identification of two comparison groups and a simple binary treatment. Reflecting the nature of the outlays' disbursement, differences across states arose from the amounts of spending received by each of them implying a continuous treatment assignment. More-

over, all US states received a considerable initial amount of funding ($\geq 10\%$ of the total outlays) in two close tranches: April 2009 (27 of them) and July 2009 (the remaining 23), none of them received no treatment. The statistical power offered by a two-months delay in treatment assignment is insignificant for the purpose of ATT estimation, thus making any distinctions between treatment and control groups purely based on funds receipt unfeasible. However, the simultaneous assignment of subsides safeguards the implemented estimates from the potential source of bias deriving from the staggered assignment[1].

In a two-way fixed effect (TWFE) basic design, the two major potential sources of endogeneity (entity peculiar characteristics and shared time trends) are controlled by the implementation of entity and time fixed effects. In this paper, other potentially biasing confounders are accounted for by the usage of the fitted values deriving from the three sector-specific instrumental variables, and an array of control variables that include: three-year moving average change in real annual personal income per capita, house price growth from 2003 to 2007, population size (Wilson, 2012) and manufacturing share of nonfarm employment (Chodorow-Reich, 2019). Capturing these endogenous components allows for a clean identification of the total amount of outlays exogenously distributed across states and thus for a reliable comparison offered by the first specification implemented. Through the usage of the basic TWFE models I compute the rather naïve comparison of the ARRA effects on employment between the most treated units of the sample against the least treated ones.

$$Y_{it} = PO_{it}\beta + X_{it}\gamma + \alpha_i + \theta_t + \epsilon_{it} \tag{1}$$

Here, Y_{it} is the employment level in state i at time t, PO_{it} is the treatment dummy classifying states with predicted outlays over the median as treated and those below this threshold untreated. X_{it} is the set of control variables, α_i and θ_t are state and time fixed effects. In a similar exercise, I also compare states sitting in the top quartile of the predicted outlays distribution vis-à-vis those belonging to the bottom one.

Although informative and helpful in setting the benchmark for the latter continuous treatment difference-in-difference estimation, this initial specification suffers from multiple perspectives. The clearest one is the treatment identification. Supposing that a state is untreated because it belongs to the lower half (or the lowest quartile) of the distribution of outlays is inaccurate and is likely to lead to an underestimation of the treatment effect. Furthermore, literature involved in this field has addressed and demonstrated the potential bias deriving from spillover effects (Dupor and McCrory, 2018; Boone et al., 2014). States that might have received less funds might still be positively contaminated by neighboring states that were highly treated, thus deflating the effect of the overall treatment. For these two reasons, I consider the reported results obtained from these two basic TWFE models the lower bounds of the treatment effects of ARRA public spending

[1]Results of Goodman-Bacon (2021) decompositions show no evidence in favor of negative weighting of treatment cohorts. Butts and Gardner (2021) method used to account for staggering treatment and negative misallocation of weights collapses to the standard TWFE result demonstrating the redundancy of these staggering methods in this setting.

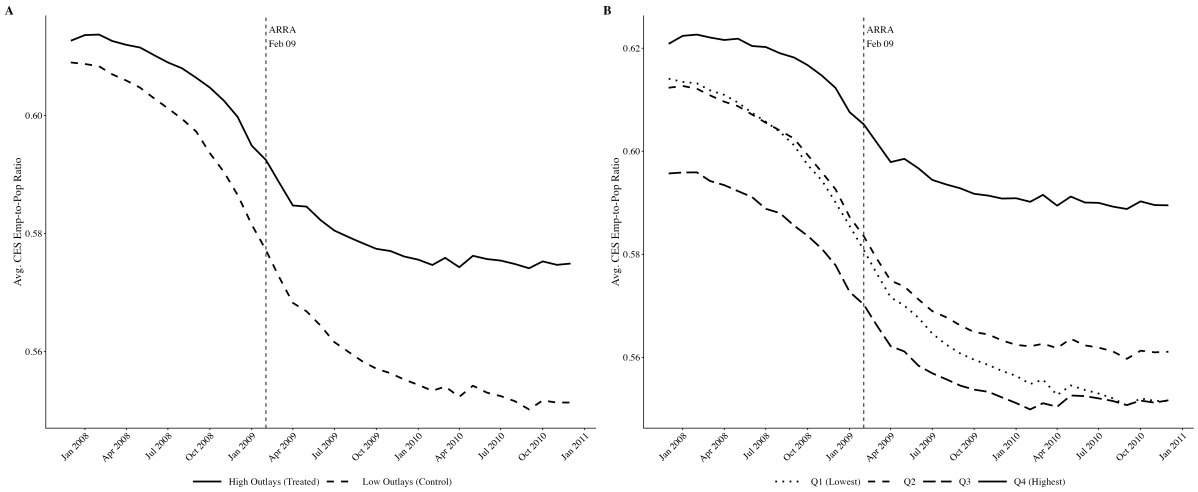

Figure 1: Parallel Trends: Median Split & Quarters

on employment.

Following the established literature on ARRA employment multipliers, I assume that anticipation effects have a minimal influence on observed post-funding outcomes (Wilson, 2012; Glick and Leduc, 2013). Between the announcement of the ARRA spending package, February 2009, and the subsequent actual payments, starting from April 2009, there is no evidence supporting a pre-treatment adjustment of employment levels (Figure 1) while subsequent treatment effects are estimated using predicted outlays which do not suffer from any form of anticipation bias.

I found, however, a positive significant estimate of pre-trend divergence in employment levels when studying both median and quartiles comparison groups. The differential slope tests yield significant estimates for both cases, showing that above mean treated states were trending at 0.0007 percentage point per month faster than controls before February 2009. Comparing lowest and highest quartile yields instead a 0.0012 percentage points difference before ARRA implementation. The inclusion of employment lags designed to absorb such pre-treatment dynamics shows that these small deviations have little impact on TWFE estimates (Table 1) but highlights the need for a more sophisticated and better tailored estimation approach[2].

2. CGBS estimator introduction and its comparison to TWFE

In continuous treatment assignment settings, the TWFE standard coefficient is shown to suffer from serious limitations and to have multiple different interpretations (Callaway et al., 2024). Defining $ATT(d|d)$ as the average effect of receiving dose d of treatment against receiving nothing and $ACRT(d|d)$ as the marginal effect of receiving an additional unit of treatment. It is shown in a TWFE coefficient decomposition that when this latter

[2] Callaway and Sant'Anna (2021) test on parallel trends conditional on covariates show no statistical significance of pre-trend diverging behavior. The statistical power of this test, however, is limited by the lack of degrees of freedom which furthermore yield insignificant estimates for the whole post-treatment periods. (see Appendix Figure 1).

is interpreted as the average level treatment effect $ATT(d|d)$, the algorithm assigns negative weights to below-mean-dose units and positive weights to above-mean-dose units, making high-dose units implicit treated and low-dose units implicit controls. When interpreted as an average causal response $ACRT(d|d)$, the TWFE include a selection bias term that standard parallel trend assumption cannot eliminate, and weights are assigned differently from the actual dose distribution.

Callaway, Goodman-Bacon and Sant'Anna estimator (onwards referred to as CGBS estimator) aim at the resolution of these issues through the implementation of:

$$ATT_{CGBS} = E[ATT(D|D)|D > 0] = E[\Delta Y|D > 0] - E[\Delta Y|D = 0] \tag{2}$$

a binarized DiD that under parallel trends allows for the identification and clean causal interpretation of the average outcome change among treated units minus average outcome change among untreated units.

$$ACRT_{CGBS} = E[ACRT(D|D)|D > 0] \tag{3}$$

where $ACRT(d|d)$ is the slope of $ATT(d|d)$ function at dose d. This derivative estimator measures the slope of the dose-response function averaged across recipients and weighted by the intensity of the dose itself.

The usage of this estimator is superior to TWFE because it directly accounts for infinitesimal differences in treatment, solving the treatment-assignment identification problem previously discussed. Furthermore, it relies on different and better tailored assumptions for this framework. Differently from the standard parallel trends assumption made in the TWFE model, CGBS estimator requires an "equal average evolution of the potential outcomes for the untreated units within same dose groups" — untreated units' evolutionary path in case of treatment should have been the same as the one of treated units within their group. By conditioning on the pre-employment slope and time-varying controls across dosage groups, the estimator absorbs the significant effects found in the differential tests and returns an insignificant pre-trend difference. Although I previously managed to account for pre-treatment slopes through linear interactions, the CGBS is the unique DiD estimator able to compute the dose-specific pre-trend variations in each period, thus being perfectly fitting for this analysis. The covariate conditioning at each dose level I performed using the CGBS also accounts for potential sources of bias correlated with both the amount of dosage received and the employment level.

Lastly, the CGBS estimator works efficiently in context in which all entities receive some degree of treatment. Evaluating how every additional unit of dosage affects the outcome within similar groups enables a more reliable comparison. The no-anticipation restriction assumed before and tackled by the usage of instrumental variables holds in this setting as well. The only remaining and relevant assumption that could still pose a threat on the newly estimated coefficients is SUTVA, potential spillover effects not yet accounted for in this setting.

3. CGBS comparison to previously adopted approaches

The implementation of CGBS is an attempt to build upon the techniques that have been used up to now in the context of the American Recovery and Reinvestment Act. While recognizing the strength of the used instruments and controls, which are kept in this paper, I try to improve the standard 2SLS estimation by breaking the cross-dimensional analysis and adapting this data to a more appropriate panel structure.

Research using instrumental variable across a cross-section of states has yielded in previous investigations a significant, valid and easy to interpret result. The fiscal job multiplier in dollar terms answers a very precise question: how many jobs have been created or saved by every spent dollar. Although relevant and perfect for policy makers, this sort of estimation is not able to i) explain the evolution of employment creation over the years ii) address heterogeneous effects across the distribution of receipts iii) estimate the weight, and thus the difference, between the first dollar spent and the last one. These three shortcomings can lead to a systemic underestimation of the effects and to a potential manipulation of the results. The cross-sectional analysis built by Wilson (2012) estimated the results of ARRA until February 2010 and estimated a cost-per-job of $125,000. A very similar analysis run by Chodorow-Reich (2019) using a cross-section of states in December 2010 estimated the effects to be less than half: $50,000. Policy makers using aggregated results can argue in favor of the reform highlighting the great effect that ARRA had on American economy, but what if similar effects could have been achieved using half of the dollars spent? What if some states were over-funded and how persistent were these public investment?

My estimation has the goal of answering these questions and reconcile the diverging literature on the fiscal job multiplier. The baseline TWFE model supported by instrumental variable implementation and strong assumptions could already solve the dimensionality issue by exploiting the panel data design. By relaxing some of these assumptions and handling the granular data, CGBS estimator is capable of returning reliable estimates of the effects of the dose-specific evolutionary trends over time. This allows to inspect the heterogeneous effects that public funding had on different cohorts (Appendix figure 2), at different time periods and for each dollar spent (Appendix Figure 3).

IV Results

Table 1 illustrates the results obtained from all the previously discussed specifications. The baseline specification is the cross-sectional analysis using IV estimator carried out by Chodorow-Reich (2019) original paper. I add three specifications using TWFE estimator and one specification reporting the estimate computed through the CGBS estimator.

An overall accordance in the sign and significance of the coefficients shows that all estimators are able to capture a positive and significant effect of ARRA spending on employment levels. As expected, the different treatment assignment threshold in TWFE models carries with it a relevant discrepancy in estimated effects. Supporting the dis-

Table 1: ARRA Employment Multiplier Estimates

Estimator	(1)	(2)	(3)	(4)	(5)
CR Baseline IV	4.27***				
	(1.28)				
TWFE Median		0.0124***	0.0123***	0.0073***	
		(0.0030)	(0.0030)	(0.0023)	
TWFE Q4 vs Q1		0.0189***	0.0190***	0.0101***	
		(0.0050)	(0.0048)	(0.0032)	
CGBS ATT					0.0182***
					(0.0054)
Instrumented	Yes	Yes	Yes	Yes	Yes
Controls	Yes	No	No	Yes	Yes
Time Trends	—	No	Yes	Yes	—

Notes: Clustered SE at state level in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

cussed evidence in favor of parallel trends, once I condition these estimates on lagged employment trends, the measured effects do not suffer from major divergence. However, once the set of control variables is included in both TWFE models, the magnitude of their estimates is halved. Interestingly enough, time and entity fixed effect are not capable of excluding all the variation deriving from confounders, suggesting that a large share of it derives from time-varying, state-specific characteristics which correlate both with treatment intensity and employment outcomes.

The last specifications represents the average treatment effect on the treated (ATT) obtained from the CGBS estimator. It is a condensed statistic of the dynamic event-study illustrated in Figure 2 which I regard to as the main contribution of the paper.

The dynamic event study shows no relevant pre-treatment divergence in potential outcomes for observations belonging to the same dosage-group and a significant increase in those outcomes after the ARRA implementation. Through faithfully weighted comparisons based on dosage reception, we are able to infer a heterogeneity-robust dynamic evolution of the employment effects caused by ARRA funding. Benchmarked to the actual outlays supplied by the federal agencies (Appendix Figure 3), the effects of the public investment plan are found to be lagging in the initial first months and distinguished by a time compounding effect that led to a continuous increase in employment levels against decreases in funding over time. These findings are aligned with the existing labor market literature that analyzes the inertia and the adjustment delay that labor market suffers after crises. Employment depression occurring after crises are observed to be persistent trends over time, both from a firm-level matching inefficiency (Davis et al., 2012) and from a macro-level adjustment sluggishness (Mota and Vasconcelos, 2020). Fiscal policies are inherently slow at targeting these malfunctioning mechanisms (Leeper et al., 2010),

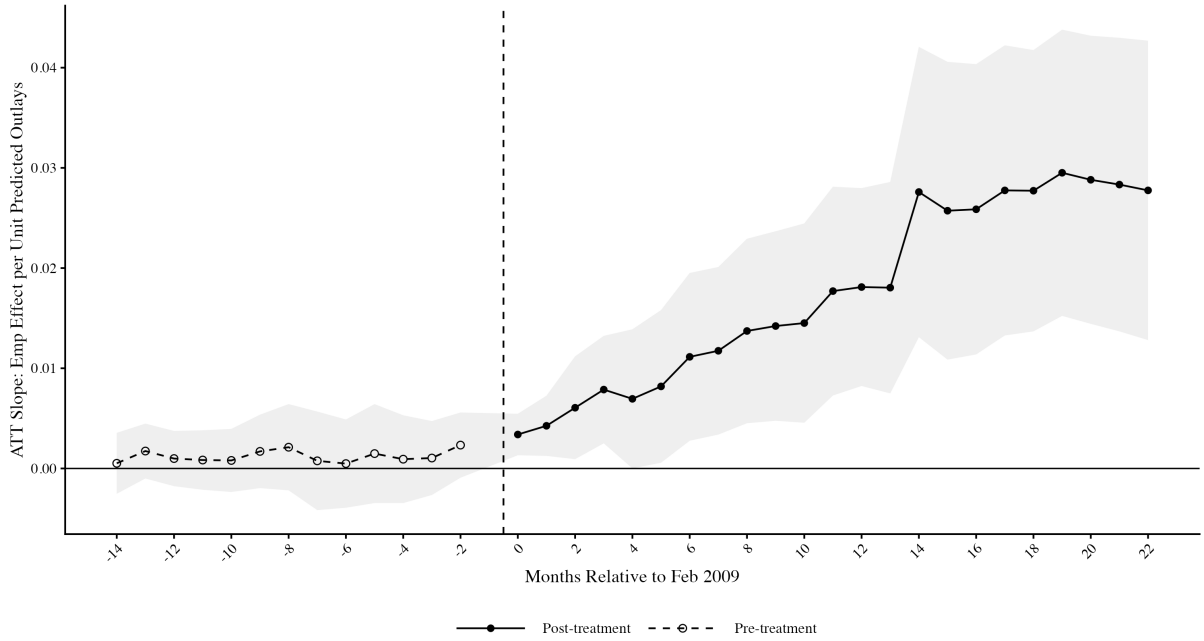

Figure 2: Dynamic Event Study: CGBS

but their effects are often observed to be compounding over time (Bouakez et al., 2020) thanks to structural increases in productivity that tend to be difficult to be implemented but long-lasting.

To compare the findings of the existing literature with the results of this study, I transform the coefficients from earlier models into a standardized metric representing the expense associated with every job generated or saved. The standard formula used to calculate the job multiplier is:

$$\$/job = \frac{\Delta outlays}{\Delta emp} \tag{4}$$

Following his own conversion, Chodorow-Reich (2019) ratio is computed as:

$$\beta_{IV} = \frac{\Delta(emp/pop) \cdot 100,000}{\Delta(outlays/pop)} \Rightarrow \$/job = \frac{10^5}{\beta_{IV}} \tag{5}$$

As β_{TWFE} estimates:

$$\frac{\Delta \text{employed workers}}{pop}, \quad \text{defining} \quad gap = \frac{outlays_{\text{treated}} - outlays_{\text{untreated}}}{pop} \tag{6}$$

I compute the Wald estimator for the TWFE models as:

$$\$/job = \frac{gap}{\beta_{TWFE}} \tag{7}$$

In the first model specification, the gap between treated and untreated units is defined by the difference in average predicted outlays between states that received above median funds and those that received below. In the second TWFE model specification this difference is instead computed between average predicted outlay in the first quartile

against the bottom one.

The conversion of the estimates obtained through the CGBS method is less straightforward and produces a unitary value of less interpretational power than the previous event-study analysis. Important to mention, the CGBS slope is estimated in standardized dose $d \in [0, 1]$:

$$d = \frac{\text{predicted_outlay} - \min(\text{predicted_outlay})}{\max(\text{predicted_outlay}) - \min(\text{predicted_outlay})} \tag{8}$$

thus, its coefficient for δ measures the difference in employment population based on standardized dose changes:

$$\delta = \frac{\Delta emp/pop}{\Delta d} \tag{9}$$

Defining

$$dose_range_dollar = [\max(\text{predicted_outlay}) - \min(\text{predicted_outlay})] \ \$/\text{capita} \tag{10}$$

I can then convert δ in dollar terms:

$$\frac{\delta}{dose_range_dollar} = \frac{\Delta emp}{\Delta outlays} \Rightarrow \$/job = \frac{dose_range_dollar}{\delta} \tag{11}$$

Which predicts the average cost of creating a job by comparing employment levels between the highest treated state and the lowest treated one. By comparing these two groups this value (reported in Table 2) measures the average costs of creating an additional job over the whole sample.

The CGBS estimate, however, allows also for comparison across groups receiving nearly identical levels of treatment. By setting a different *dose_range_dollar*, it is possible to measure the heterogeneous effect that ARRA spending had on states sitting at any point of the distribution and separated by any distance (in treatment levels).

Table 2: Cost per Job Multipliers

Estimator	β	Cost per Job
CR Baseline IV	4.27***	\$23,419
	(1.28)	
TWFE Median	0.0073***	\$23,719
	(0.0023)	
TWFE Q4 vs Q1	0.0101***	\$37,360
	(0.0032)	
CGBS ATT	0.0182***	\$46,223
	(0.0054)	

V Conclusion

In the reassessment of the American Recovery and Reinvestment Act spending effects on employment levels my estimation has found consistent results that, while in sharp contrast with the earliest literature, align more closely to the most recent results. Wilson (2012), found an average cost per job created or saved of about $125,000. In a similar exercise, Feyrer and Sacerdote (2011) found an even higher cost of roughly $170,000 per job. These results, such as the initial projection that forecasted a cost per job of around $92,000 (Romer and Bernstein, 2009), are considered an under-estimation of the true effect of ARRA spending by the most contemporaneous literature. Other than Chodorow-Reich (2019) estimate of $23,000, from which the author builds a more reliable interval between $20,000 to $50,000, my results are in accordance with the findings of Boone et al. (2014), reporting a figure of $25,000 per job and those of Choi et al. (2024), who, focusing their primary analysis on political connections and instrumenting the party ruling each state, approximate the costs per job at $37,000.

Through the introduction of new estimation techniques, this analysis was able to improve the existing literature in two major directions. By firstly introducing two specifications that exploit the panel data structure of the data and account for time trends, I was able to break the cross-sectional constraint that forced previous studies to estimate results on a single dimensional cumulative distribution. By a latter implementation of a continuous treatment effect estimator, I was able to address questions related to the heterogeneous effects of the treatment over time and across cohorts, observing an average cost per job created or saved of about $46,000. By providing policy makers new instruments to evaluate the impact of each public dollar invested and the accuracy of the targeting techniques, these novel contributions pave the way for a new relevant policy discussion.

However, this analysis is not immune to potential limitations. For the previously mentioned reasons, both TWFE resulting coefficients must be handled with caution and only used for the purpose of benchmarking. The latter CGBS estimator, although highly sophisticated and perfectly matching the analysis's objective and assumptions, is quite new on the econometric literature. Firstly published in 2024, it has been revised in 2025 and updated in 2026, but it has not yet been challenged or criticized by the econometric community. By missing the peer-to-peer review process, its derivations and results cannot be taken for granted and definitively validated.

While inviting future research to apply such method to other frameworks that involve public funding, I urge to indicate two additional possible avenues that are worth mentioning to improve this paper's results. The potential for SUTVA violations and varying speed of investment expenditure represent challenges that may be addressed by applying micro-founded assessment across various sectors impacted by the ARRA. By looking at public projects, firm-level developments and changes in consumer choices due to tax cuts, the analysis can be enriched and remarkably improved.

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Appendix

This section contains supplementary figures detailing the actual outlays supplied by federal agencies and the heterogeneous effects across different dosage cohorts.

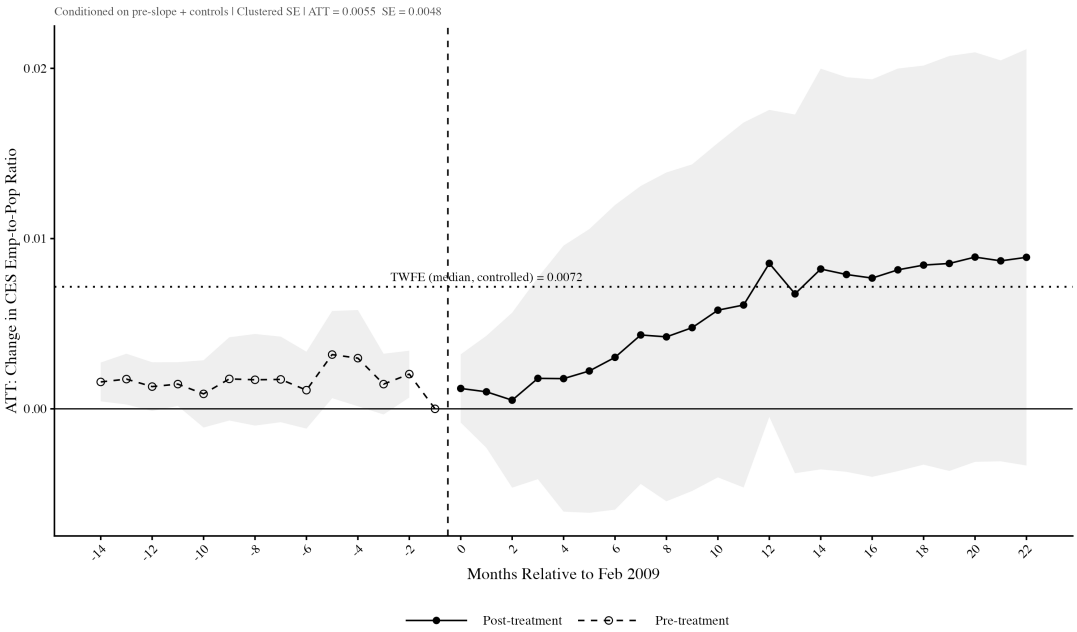

Figure 1: Callaway & Sant'Anna Event Study, Q4 v. Q1

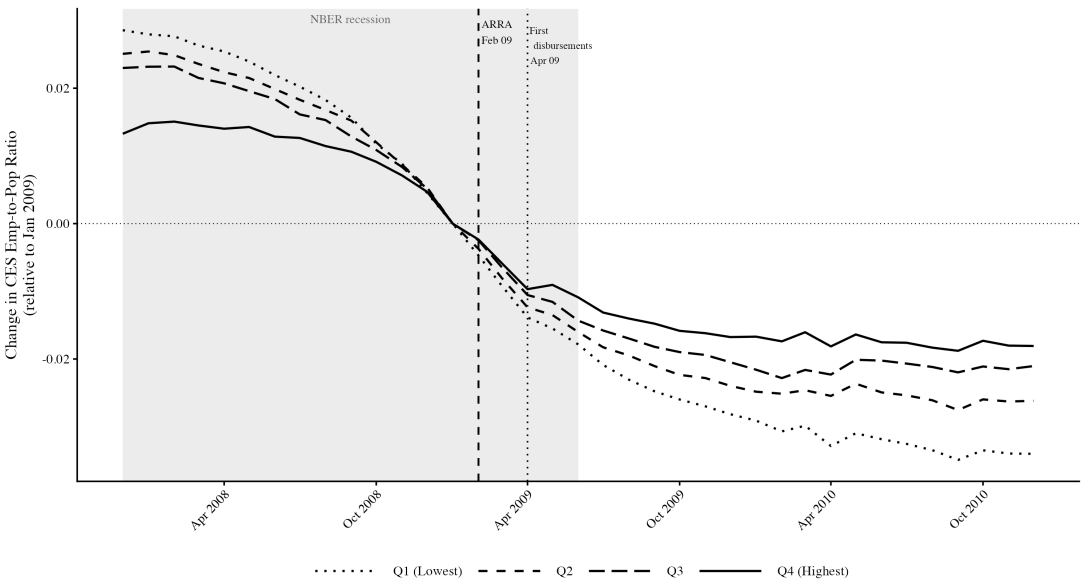

Figure 2: Employment Recovery by Stimulus Quartile

Notes: In an exercise aimed at studying the different stimulus effects across quartiles, I show the heterogenous effect that ARRA spending produces across the population. Intuitively, states suffering the most from the crisis were the ones receiving highest treatment and afterwards on moving on steeper recovery slopes.

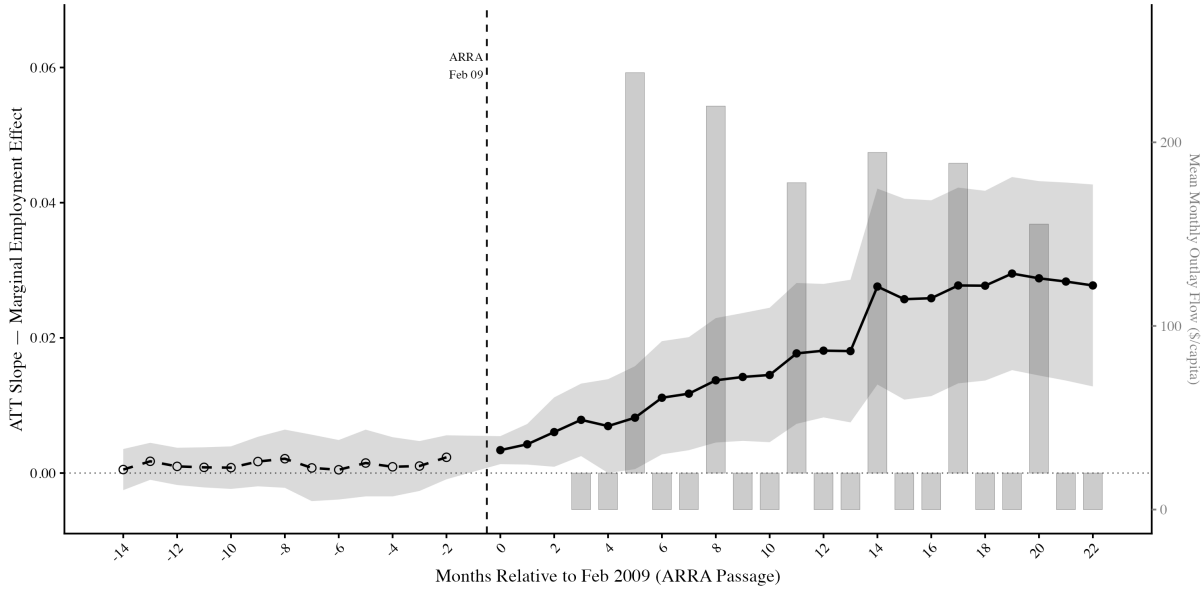

Figure 3: ARRA Spending Pace v. Marginal Employment Effect

Notes: In order to measure the average impact that each dollar of stimulus produced overtime, I plot the average continuous treatment effect obtained from CGBS estimator and the mean monthly outlay granted by ARRA legislation. Although difficult to assess due to high persistence in the labor market and lagged effect of employment upon receiving funds, it is reasonable to conclude that changes in employment followed spending receipt in an overall proportional way, showing no evidence in favor of decreasing marginal effects of spending.